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Forecasting the Volatility of US Oil and Gas Firms With Machine Learning

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ABSTRACT

Forecasting the realized volatility of oil and gas firms is of interest to investors and practitioners trading on the energy spot and derivative markets. In this paper, we assess whether several machine learning (ML) techniques can offer superior forecasts compared to HAR models for predicting realized volatility at the firm level. Moreover, we investigate whether economically motivated variables and technical indicators contain valuable information for forecasting firm volatility beyond those contained in various volatility factors previously identified in the literature. Our results demonstrate that certain ML techniques provide superior forecasting accuracy compared to the benchmark model. Additionally, we identify variables such as the 1-month treasury bill and the aggregate VIX index as significant drivers of realized firm volatility in the oil and gas industry.

1 | Introduction

Energy price shocks have sizeable impacts on financial markets. In this context, forecasting the volatility of oil and gas firms is of great interest to investors and practitioners trading in the energy spot and derivative markets (see, e.g., Antonakakis et al. 2018; Kang, de Gracia, and Ratti 2017). For example, in a recent study, Lyócsa and Todorova (2021) investigate the volatility drivers of US oil and gas firms listed on the S&P 500 index between 2007 and 2017. The authors focus on the role of volatility factors. More specifically, they evaluate the predictive power of each firm's own volatility, industry market volatility, local (US) market volatility, world equity market volatility, oil price volatility, and natural gas price volatility. Their study concludes that the main drivers of firm volatility are market volatility, each firm's own volatility, and industry level volatility. Moreover, the authors find that by using a HAR model incorporating all the factors, they can make out-of-sample forecasts with 4 % fewer errors than their benchmark model, that is, Corsi's (2009) standard HAR model.

As is customary in the literature, the volatility forecasting exercise uses a linear framework and the ordinary least squares (OLS) estimation method. In this study, we relax this assumption and investigate to what extent more flexible machine learning (ML) methods improve the accuracy of US oil and gas firm realized volatility (RV) forecasts. Recent literature have shown that ML methods improve the RV forecasting accuracy at the firm level (see, e.g., Christensen, Siggaard, and Veliyev 2023; Li and Tang 2021; Rahimikia and Poon 2020; Audrino, Sigrist, and Ballinari 2020; Zhu et al. 2023) and at the aggregate level (see, e.g., Mittnik, Robinzonov, and Spindler 2015; Bucci 2020; Díaz, Hansen, and Cabrera 2024; Zhang, Ma, and Wang 2019; Zhang et al. 2019; Dai, Li, and Yang 2022). Thus, we study whether we observe similar forecasting gains for firms in the oil and gas industry.

Furthermore, we study whether macro-finance variables and technical indicators provide additional predicting gains beyond those variables generated by volatility factors identified as the main drivers of volatility in this industry by Lyócsa and Todorova's (2021).

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The study of macroeconomic variables can theoretically be motivated by a two-factor model similar to the one proposed by Engle, Ghysels, and Sohn (2013). This framework models volatility dynamics through a short-run component, capturing daily fluctuations assumed to be short-lived, and a long-run component, reflecting the long-run state of the economy indicated by macroeconomic variables. Moreover, extensive empirical literature demonstrates that both macroeconomic and financial variables serve as valuable predictors of volatility (see, e.g., Christiansen, Schmeling, and Schrimpf 2012; Engle, Ghysels, and Sohn 2013; Nonejad 2017; Christensen, Siggaard, and Veliyev 2023; Zhu et al. 2023; Asgharian, Christiansen, and Hou 2023).

The inclusion of technical indicators is also supported by prior empirical literature, which suggests that technical indicators such as moving averages (MA) and momentum (MOM) indicators have the potential to enhance forecasts of both stock market volatility (see, e.g., Díaz, Hansen, and Cabrera 2024; Liu and Pan 2020) and oil price volatility (Ma et al. 2018).[1] Hence, this paper assesses to what extent additional economic variables and technical indicators may yield more accurate forecasts of the RV of US oil and gas firms compared to just volatility factors.

In recent years, ML models have been increasingly leveraged in economic and financial forecasting exercises.[2] For example, Gu, Kelly, and Xiu (2020) evaluates several ML techniques' ability to estimate stock risk premia in the United States and Bianchi, Büchner, and Tamoni (2021) study bond risk premia in the same country. In the same vein, recent research has evaluated the use of ML techniques in volatility forecasting exercises at the aggregate level (see, e.g., Mittnik, Robinzonov, and Spindler 2015; Díaz, Hansen, and Cabrera 2024; Bucci 2020) and at the firm level (see, e.g., Christensen, Siggaard, and Veliyev 2023; Li and Tang 2021; Rahimikia and Poon 2020). Guided by the models used in these applications, we examine whether regularization methods (lasso, ridge, and elastic net) and tree-based methods (random forest [RF] and gradient boosting) can improve the accuracy of US oil and gas firm RV forecasts. Initially, we estimate the ML models using the original set of predictors presented by Lyócsa and Todorova (2021). Subsequently, we expand this set by incorporating additional economic variables and technical indicators.

First, we show that ML methods can yield more accurate forecasts of oil and gas firm volatility than the HAR model with the set of predictors presented by Lyócsa and Todorova (2021) (the benchmark model). As such, we provide evidence that ML methods are useful for forecasting the volatility of firms in the US oil and gas industry. Estimated forecasting gains vary according to firm and ML method, but they are as high as 5%. Furthermore, when using the MCS test of Hansen, Lunde, and Nason (2011) to identify the set of best forecasting models, we find that several ML methods almost always belong to this optimal set, whereas the benchmark model does not. Our findings reveal that a forecast combination of the five ML methods is the top performing model of firm volatility, followed by the Ridge method, which also performs well across forecasting exercises.

Second, we show that, besides the volatility factors, other economic variables contain valuable information with which to forecast firm volatility. We find that the US 1-month treasury bill and the option-implied VIX index are particularly important in this regard. Additionally, while assessing the predictive capacity of several technical indicators in forecasting firm volatility, we observe that these variables do not exhibit notably high predictive power.

The remainder of this paper is structured as follows. In Section 2, we describe our dataset; in Section 3, we describe the empirical setting including the ML methods evaluated, cross-validation details, and forecasting framework. We report our baseline empirical results in Section 4. In Section 5, we provide additional empirical analysis; and we conclude in Section 6.

2 | Data

This paper uses an extended version of the data from Lyócsa and Todorova (2021), which are available online on the website of the Energy Economics journal. Data are available for 15 S&P 500 companies belonging to the sub-industry of oil and gas exploration and production. They cover the period between January 2007 and December 2017 and their frequency is daily. The main variable of interest here is each firm's RV (based on the sample), which was computed using intraday prices. More specifically, the RV of trading day t is given by $RV^2_{t,5m} = \sum_{j=1}^{N_{t,5m}} r^2_{t,j,5m}$, where $r^2_{t,j,5m}$ is the continuous return within a 5-min frequency and $N_{t,5m}$ is the number of returns given the sample frequency. Following a suggestion by Patton and Sheppard (2009), the RV of trading day t is computed as the average RVs at 5, 10, 15 and 30 min: $RV^2_t = \left(RV^2_{t,5m} + RV^2_{t,10m} + RV^2_{t,15m} + RV^2_{t,30m}\right)/4$. Finally, firms' RV are adjusted by incorporating overnight price variation in their definition, as follows: $RV_t = \ln\left(RV^2_t + OV^2_t\right)$, where OV^2_t is an overnight volatility estimate.

We start estimating the volatility factors of the Corsi's HAR model (2009), where weekly and monthly RVs are defined as $RV^W_t = \left(\frac{1}{5}\right)\sum_{t=0}^{4} RV_{t-i}$ and $RV^M_t = \left(\frac{1}{22}\right)\sum_{t=0}^{21} RV_{t-i}$, respectively. Then, we use the same six volatility factors employed by Lyócsa and Todorova (2021) as predictors. These predictors are each firm's RV, the industry-level RV factor, the RV of the US equity market estimated using S&P mini-index futures, a world equity market volatility factor estimated using MSCI world market index data, an oil volatility factor estimated using WTI future contracts, and finally, a natural gas volatility factor estimated using Henry Hub futures contracts.

We extend this set of predictors by adding economic variables. We include the CBOE option-implied volatility (VIX) index, the US 1-month T-bill rate, the Aruoba, Diebold, and Scotti (2009) business conditions (ADS) index, and the Economic Policy Uncertainty index from Baker, Bloom, and Davis (2016). These four variables are selected due to recent evidence from Christensen, Siggaard, and Veliyev (2023) that they can help predict the RV of US stocks at daily frequencies. All these variables are retrieved from the Federal Reserve Bank of St. Louis' (FRED) online database, apart from the ADS index, which is sourced from the Federal Reserve Bank of Philadelphia's webpage.

Finally, for each firm in the sample, we include signals defined from a set of technical indicators and technical rules. We consider two moving average (MA) indicators, two momentum (MOM) indicators and two volatility clustering (VOA) indicators. Each technical indicator defines a signal (S) that takes a value of 1 or 0 for long and short positions, respectively. The specific definitions of this set of signals are as follows. A moving average signal is given by

$$S_{i,t} = \begin{cases} 1 \; if \; MA_{s,t} \geq M_{l,t}, \\ 0 \; if \; MA_{s,t} < MA_{l,t}, \end{cases}$$

where

$$MA_{j,t} = \left(\frac{1}{j}\right) \sum_{i=0}^{j-1} P_{t-i},$$

for $j = s, l$, where P_t is the asset price level, and $s(l)$ is the length of the short (long) MA ($s < l$). We considered MA rules with $s = 1$ and $l = 5, 22$.

The momentum strategy $MOM(m)$ defines the following signal:

$$S_{i,t} = \begin{cases} 1 \; if \; P_t \geq P_{t-m}, \\ 0 \; if \; P_t < P_{t-m}. \end{cases}$$

We evaluate $m = 5, 22$. Finally, the volatility clustering indicators are

$$S_t = \begin{cases} 1 \; if \; RV_t \geq RV_{t-k}, \\ 0 \; if \; RV_t < RV_{t-k}, \end{cases}$$

where RV_t is the daily RV during day t. We consider $k = 5$ and 22.[3] Table A1 shows the descriptive statistics of the volatility factors and economic variables used as predictors in the empirical analysis. Table A2 shows the correlation among these variables.[4]

3 | Empirical Setting

This section presents the benchmark model and the ML methods employed to forecast the RV of oil and gas firms, including regularization and tree-based methods. Thereafter, we provide details of the forecasting exercise and the formal testing of differences in forecasting accuracy among ML methods.

3.1 | The Benchmark Model

We use as benchmark model throughout the study the best out-of-sample forecasting performing model identified in Lyócsa and Todorova (2021) for the set of oil and gas firms used in their study. Specifically, the model is a HAR model (Corsi 2009) expanded with a set of five additional factors. The model is named the all-factor HAR (AF-HAR) model. The model specification is the following:

$$\begin{aligned} RV_{t+1} = \beta_0 + \beta_1 RV_t^D + \beta_2 IRV_t^D + \beta_3 MRV_t^D + \beta_4 WRV_t^D \\ + \beta_5 ORV_t^D + \beta_6 GRV_t^D + \beta_7 RV_t^W + \beta_8 RV_t^M + \nu_t, \end{aligned}$$

where RV_t^D, RV_t^W, and RV_t^M are daily, weekly, and monthly RV of the firm, respectively; IRV_t^D is the industry volatility; MRV_t^D is the US market volatility; WRV_t^D is the world market volatility; ORV_t^D is the oil market volatility; and GRV_t^D is the gas market volatility.[5]

3.2 | ML Methods

3.2.1 | Regularization Methods

We consider three types of penalized linear regression: ridge (Hoerl and Kennard 1970), lasso (Tibshirani 1996), and elastic net (Zou and Hastie 2005). These methods build on the classic linear regression, but the coefficient estimates are obtained by minimizing a penalized residual sum of squares. Specifically, consider the following model:

$$\boldsymbol{R} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{1}$$

where $\boldsymbol{R}$ is a $T \times 1$ vector that includes the outcome variable of interest, where T is the number of observations used to estimate the model; $\boldsymbol{X}$ is a $T \times p$ matrix comprising the predictors included in the model, where p is the number of predictors included in the model; $\boldsymbol{\beta}$ is a $p \times 1$ vector of coefficients to be estimated; and $\boldsymbol{\epsilon}$ is the $T \times 1$ vector of error terms, each one independent and identically distributed (i.i.d.), with a mean of zero and constant variance. The elastic net method estimates $\boldsymbol{\beta}$ by solving the following optimization problem:

$$\min_{\beta} \|\boldsymbol{R} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \cdot \left(\alpha \cdot \|\boldsymbol{\beta}\|_1 + (1-\alpha) \cdot \|\boldsymbol{\beta}\|_2^2\right) \tag{2}$$

for $\lambda > 0$ and $0 \leq \alpha \leq 1$, where $\|\cdot\|_1$ is the l^1 norm and where $\|\cdot\|_2$ is the Euclidean norm. In this setting, the tuning parameters λ and α are selected using a time-series cross-validation procedure. This procedure is described in Section 3.5. The grids of values for these parameters tested in this cross-validation procedure is presented in Table 1. These grids of values are used by Christensen, Siggaard, and Veliyev (2023).

It is important to note that if we set $\alpha = 0$, then the optimization problem in (2) corresponds to that of a Ridge regression, while if we set $\alpha = 1$, it corresponds to that of the Lasso method. Regardless of whether $\alpha = 0$ or $\alpha = 1$, the tuning parameter λ is selected using a time-series cross-validation procedure. This procedure is described in Section 3.5 and the grid of values for this parameter tested in this cross-validation procedure is presented in Table 1.

3.2.2 | Tree-Based Methods

We consider two ML methods based on trees: the RF (Ho 1995; Amit and Geman 1997; Breiman 2001) and gradient boosting regression tree (GBRT, Hastie, Tibshirani, and Friedman 2009, Chapter 10).

TABLE 1 | Tuning of hyperparameters.

Regularization	RF	GBRT
$\lambda \in [10^{-5}, 10^{2}]$*	Min. Node size = 5	Depth $\in \{1, 2\}$*
$\alpha \in [0, 1]$*	Trees = 500	Trees $\in \{50,100, \ldots, 500\}$*
	Feature split = $J/3$	Learning rate $\in \{0.01,0.1\}$*

Note: The table provides an overview of the hyperparameters for the machine learning algorithms based on Christensen, Siggaard, and Veliyev (2023). An asterisk means that the hyperparameters undergo tuning in the validation set. For tuning λ, we perform grid searching over the natural logarithm across an interval defined by a partition consisting of 1000 points. The hyperparameter α is defined by a partition consisting of 11 points. RF (random forest): The parameters listed are defaults derived from Breiman and Cutler (2005), where J denotes the number of predictors. GBRT (gradient boosting regression tree): The parameters listed are defaults based on the implementation by Greenwell et al. (2019).

Both methods are ensemble statistical methods that extend the single regression tree (Breiman et al. 1984) by combining predictions from multiple regression trees. Specifically, RFs exploit the idea of bootstrap aggregation (bagging, Breiman 1996) by computing their final prediction as the average of those from several single regression trees. These single regression trees are adjusted with random subsets of training data and predictors. Meanwhile, based on the idea of boosting (Freund and Schapire 1996; Schapire 2003), GBRTs provide better prediction accuracy than single regression trees by sequentially decreasing the bias of simple regression trees with higher bias but low variance, where the contemporary ensemble regression tree model internalizes the knowledge acquired from previously fitted models.

In practice, these methods have tuning parameters that must be set. For the RF, we choose to randomly select $J/3$ predictors at each split, where J denotes the number of predictors; we employ 500 regression trees to be aggregated; and we estimate trees with a minimum node size of five observations (their complexity parameter). The parameters listed are defaults derived from Breiman and Cutler (2005) and Christensen, Siggaard, and Veliyev (2023).

For the GBRT, we determine the number of trees to be aggregated in the final model, the depth, and the learning rate using a time-series cross-validation procedure. This procedure is described in Section 3.5. The grids of values for these parameters tested in this cross-validation procedure is presented in Table 1 and are used by Greenwell et al. (2019) and Christensen, Siggaard, and Veliyev (2021).

3.3 | Forecast Combination

Prior literature has shown that forecast combinations can improve the accuracy of forecasts (see, e.g., Rapach, Strauss, and Zhou 2010, and the authors' references). Therefore, we perform two exercises to obtain predictions from forecast combinations. First, following Zhang et al. (2020), we combine forecasts from specific ML models with those from the benchmark model. Zhang et al. (2020) show that in the case of aggregate stock return predictability, this approach outperforms individual ML methods. In the second exercise, all the ML methods' forecasts are combined. We use equal weights in both exercises.

3.4 | Forecasting Framework

Our forecasting procedure aims to resemble Lyócsa and Todorova's (2021) to provide as precise a comparison as possible of their best model, our benchmark, and the alternative ML methods leveraging additional predictors. Specifically, we use a rolling window estimation procedure with an estimation window of 1000 observations. Thus, we estimate each model in a particular window and forecast one period ahead. Then, we move the window 1 day forward and predict 1 day again. We repeat this procedure until we obtain a forecast for the last observation in our sample.

As mentioned above, we use a sample period that extends from January 2007 until December 2017. To simulate a real out-of-sample forecasting exercise, we split the sample into estimation and testing subsamples and use the former to adjust the models and the latter to assess their predictive accuracy. We begin our empirical analysis by using an estimation subsample comprising the first 1000 days (from January 2007). The testing subsample begins on day 1001 and ends in December 2017.

The rolling window approach uses all the information in a window to train and estimate the models. The estimated models are then used to compute each new prediction in the testing sample. For example, to forecast the first day in the testing sample, we begin by fitting the model using the initial estimation sample (i.e., all the information prior to that first day in the testing sample). Then, we roll the estimation sample forward by 1 day to predict the second day in the testing sample, re-estimate the models, and predict for day two. We proceed in this manner until we obtain forecasts for all the days in the testing sample.

3.5 | Cross-Validation of the Hyperparameters

We use a time-series cross-validation procedure to tune the hyperparameters for the regularization methods, RFs, and GBRT (Christensen, Siggaard, and Veliyev 2023; DeMiguel et al. 2023). We implement this cross-validation procedure to each ML model for every firm. Table 1 presents the grids of values for the hyperparameters tested in cross-validation procedure for the ML methods.

As mentioned, the size of the estimation window is set at 1000 observations. We split this estimation window into training (750

observations) and validation (250 observations) subsamples, while respecting the temporal hierarchy of time series data. We employ the training subsample to adjust the models using different values of the hyperparameters, and the validation subsample to assess their predictive accuracy based on the Mean Squared Error (MSE). In each ML model for every firm, we use the values of the hyperparameters that minimizes the MSE for fitting the model with the estimation window and predicting the next observation in the testing sample.

Given a set of hyperparameters, we update the models' estimation each time after rolling the estimation sample forward by 1 day. We update the values of the hyperparameters after every 180 new predictions in the testing sample in line with Gu, Kelly, and Xiu (2020) and Díaz, Hansen, and Cabrera (2023) who refresh the hyperparameters after each year due to computational time constraints. For example, we begin our empirical analysis by using an estimation subsample comprising the first 1000 days (from January 2007) to set the initial values of the hyperparameters. Based on these values of the hyperparameters, we fit the models with the initial estimation window and forecast the first day in the testing sample. Then, we roll the estimation sample forward by 1 day to predict the second day in the testing sample, re-estimate the models (using the same values of the hyperparameters), and predict for Day 2. We proceed in this manner until we obtain forecasts for 180 days in the testing sample. At that period, we update the values of the hyperparameters based on the time-series cross-validation procedure. We proceed in this manner until we obtain forecasts for all the days in the testing sample.

3.6 | Metrics to Assess the Accuracy of the Forecasts

The accuracy of the forecasts is mainly evaluated using the Q-LIKE forecasting loss functions, as suggested by Patton (2011). In the robustness section, we also present results using the mean-squared forecasting error (MSFE) loss function. These functions are defined as follows:

$$QLIKE_t = \frac{RV_t}{RV_t^F} - ln\left(\frac{RV_t}{RV_t^F}\right) - 1,$$

$$MSFE = \frac{1}{M}\sum_{t=1}^{M}\left(RV_t - RV_t^F\right)^2.$$

Here, RV_t is the observed firm RV at time t and RV_t^F is the forecast from the model under evaluation.

We evaluate and compare the forecasting performance of the alternative models using the model confidence set (MCS) test of Hansen, Lunde, and Nason (2011). The test works sequentially by eliminating the worst-performing model among a set of M models until the only remaining models are those for which the null hypothesis of equal forecasting accuracy cannot be rejected at a significant level (α). The MCS contains the best performing models from M at the $(1-\alpha)$ confidence level. We set $\alpha = 0.1$ and use the T_R statistics (see details in Hansen, Lunde, and Nason 2011) with 5000 bootstrap simulations.

4 | Empirical Results

This section reports our empirical results. We report a first set of results obtained using the same predictors leveraged by Lyócsa and Todorova (2021). This allows us to quantify forecasting gains associated with the use of ML methods rather than linear least squared methods, i.e., it permits an assessment of the extent to which non-linear models can specifically improve forecasting accuracy. Subsequently, we expand the set of predictors and investigate their ability to improve forecasting accuracy. We considered two sets of additional predictors: economic variables and technical indicators. A detailed list of these variables is presented in the data section above. Thus, after reporting baseline results using Lyócsa and Todorova's predictors (2021), in the second set of results, we report findings based on the inclusion of the additional economic variables and technical indicators as predictors. In the appendix, we also report supplementary results, including those obtained following the independent incorporation of the economic variables on the one hand and the technical indicators on the other.

4.1 | Baseline Results: The Role of ML Models

Table 2 reports a set of ML methods' performance at predicting the RV of 15 firms in the oil and gas industry. Each column reports the results for a given firm. We use the Q-LIKE loss function to compare the models. The benchmark model is the "AF-HAR" model described in Section 3.1. This model was identified as the best performer among the volatility factors in Lyócsa and Todorova (2021).

For each firm in the sample, Table 2 reports the percentage of relative gains/losses in forecasting accuracy compared with the benchmark model (AF-HAR). A negative value indicates that the evaluated model produces a lower value on the Q-LIKE loss function and therefore that it delivers superior performance as compared to the benchmark model. Conversely, a positive value indicates a higher realization of the loss function and thus performance that is worse than that of the benchmark model.

After reporting the benchmark in the first row of the table, we report the relative performance of the ML methods, that is, the ridge, lasso, elastic net, RF, and Gradient Boosting Methods (GBM). Below this, we report the simple equally weighted combination of the benchmark and ML method and label it as "AF-HAR +." Finally, the last row of the table reports the equally weighted forecast combination of the five ML methods ("pooled-avg.").

Table 2 shows that some of the ML methods under evaluation offer superior performance when considering the entire sample of oil and gas firms. When considering the ML methods' performance independently, we observe forecasting accuracy gains from the regularization methods, but do not necessarily observe likewise from the tree-based methods. For example, in the case of the APA firm, as compared with the benchmark model, the Q-LIKE loss function is reduced by 0.78% when a Ridge estimator is used for the forecast. Yet, when a RF is used instead, we observe that the estimated loss function increases by 1.15% as compared to the benchmark. Similar results are obtained for the GBM method, since the loss function increases by a significant 3.17%. For most other firms in the sample, we observe a similar pattern, whereby regularization methods, particularly the Ridge method,

TABLE 2 | Realized volatility forecasting accuracy of machine learning methods. (Predictors: original volatility factors; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
AF-HAR	0.444	0.432*	0.410	0.432	0.570*	0.385*	0.415*	0.455*	0.435*	0.461*	0.427*	0.450*	0.368	0.412*	0.349*
Ridge	−0.78%*	−1.01%*	−1.02%*	1.74%	−0.73%*	−0.19%*	−0.70%*	−0.65%*	−0.70%*	−0.99%*	−0.17%*	−0.21%*	−0.89%*	−0.23%*	−0.45%*
Lasso	0.16%	−0.69%*	−0.55%*	2.28%	−0.24%*	−0.12%*	−0.19%*	−0.27%*	−0.18%*	−0.65%*	0.11%*	0.47%	−0.15%	0.78%	−0.48%*
Enet	−0.62%*	−0.92%*	−0.57%*	2.19%	−0.68%*	−0.08%*	−0.25%*	−0.52%*	−0.35%*	−0.80%*	0.14%*	0.54%	−0.83%*	0.19%*	−0.65%*
Rf	1.15%	1.74%	3.44%	0.10%	4.79%	4.89%	4.39%	3.14%	4.89%	5.90%	3.11%	−0.89%*	4.70%	1.89%	7.35%
gbm	3.17%	−0.39%*	2.28%	1.65%	0.21%*	0.95%	4.04%	1.99%	2.59%	1.96%	3.83%	−2.35%*	1.29%	2.30%	3.72%
af-har + ridge	−0.59%*	−0.65%*	−0.66%*	0.44%	−0.53%*	−0.31%*	−0.57%*	−0.61%*	−0.57%*	−0.77%*	−0.53%*	−0.38%*	−0.74%*	−0.31%*	−0.53%*
af-har + lasso	−0.09%	−0.48%*	−0.37%*	0.68%	−0.26%*	−0.18%*	−0.22%*	−0.34%*	−0.23%*	−0.55%*	−0.12%*	−0.14%*	−0.33%	0.19%	−0.45%*
af-har + enet	−0.51%*	−0.67%*	−0.41%*	0.63%	−0.54%*	−0.23%*	−0.34%*	−0.47%*	−0.35%*	−0.62%*	−0.38%*	−0.18%*	−0.72%*	−0.08%	−0.57%*
af-har + rf	−2.02%*	−1.61%*	−0.66%*	−2.41%*	−0.26%	−0.21%*	−0.16%*	−0.96%*	0.18%*	0.47%	−1.14%*	−3.10%*	0.27%	−1.17%*	1.42%
af-har + gbm	−0.60%	−2.05%*	−0.41%	−1.19%*	−1.70%*	−1.10%*	0.03%*	−1.08%*	−0.26%*	−0.62%*	−0.15%*	−3.13%*	−0.82%*	−0.45%*	0.51%*
pooled-avg	−1.79%*	−2.33%*	−1.34%*	−0.76%	−1.51%*	−0.92%*	−0.68%*	−1.62%*	−0.64%*	−0.89%*	−1.03%*	−2.82%*	−0.89%*	−0.94%*	0.15%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR +" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

outperform the benchmark in most of the cases. The ridge method performs the best, followed by the elastic net and lasso methods. The tree-based methods tend to produce poorer forecasting results than the benchmark model. The RF model yields the worst performance, while the GBM estimates are slightly better given that the benchmark is beaten in four out of 15 cases.

We observe better results when the individual ML methods are combined with the benchmark model. Indeed, in most of the cases, the forecast combinations reduce the loss function estimate, indicating more accurate predictions than the benchmark model. The gain in forecasting accuracy varies between 0.1% and 3.3% depending on the given ML method and firm. We find that the best performing ML methods are the tree-based regression methods. Specifically, for 11 out of 15 firms, the GBM method is the one that produces the most accurate forecasts when combined with the benchmark. As compared to the benchmark model, the average gain across firms from using the GBM is 0.87%. With an average gain of 0.76%, the RF method ranks as the second best. There are firms for which these relative improvements vary between 2% and 3%, making them comparable in magnitude to the original gains reported in the forecasting exercise conducted by Lyócsa and Todorova (2021). Finally, when all the ML are equally combined, the forecasting performance improved further.

As might have been expected given the performance of the forecast combination incorporating individual ML methods and the benchmark model, the performance of the forecast combination using the five ML methods simultaneously is even better. The average forecasting gain across firms is 1.2%, the highest value across all the alternatives considered up until now.

The MCS test confirms the prior findings. First, we find that for 11 of the firms in the sample, the benchmark model (AF-HAR) belongs to the optimal set of forecast models. According to the MCS test, among the ML methods we evaluated, we find that the Ridge method belongs to the optimal set all the cases but one. This result remains when combined with the benchmark model. The forecast combination incorporating the tree-based models and the benchmark belongs to this set in 11 cases. Finally, the forecast combination including the five ML methods belongs to the optimal confidence set in 14 firms in the sample.

Figure 1 shows a heat map identifying the relative importance of each variable by ML method and firm. A dark (light) red color indicates more (less) importance in the estimation of the forecasting model.[6] Similarly, to Lyócsa and Todorova (2021), we observe that among the volatility variables, the most important ones are the HAR factors (firms' own lagged volatility [rv_l1], lagged weekly volatility [rv_l5], lagged monthly volatility [rv_l22], and the industry volatility factor [irv_l1]).

Overall, we have shown so far that ML methods, either individually or combined, produce superior forecasts of firms' RV than the benchmark model. It is worth noting that our results were obtained using the same dataset that Lyócsa and Todorova (2021) leveraged, so they cannot be explained by differences in data sources or sample coverage. Furthermore, our results are robust to the use of alternative loss functions (see Section 5.1). In terms of methods, we find that the Ridge method, and the forecast combinations produce superior forecasts than competing methods.

4.2 | Additional Results: The Role of Economic Variables and Technical Indicators

This section reports results for the same set of ML methods as the previous section. Yet, instead of only using Lyócsa and Todorova's (2021) AF-HAR model containing volatility factors, we incorporate an additional set of predictors.

We start by adding a set of macroeconomic variables, motivated by prior evidence showing that they can predict stock volatility evidence (e.g. Christensen, Siggaard, and Veliyev 2023; Díaz, Hansen, and Cabrera 2023). Besides, Engle, Ghysels, and Sohn (2013) provides a theoretical framing to support this idea. In particular, these authors model unexpected returns as $r_{i,t} - E_{i-1,t}\left(r_{i,t}\right) = \sqrt{\tau_t \times g_{i,t}}\epsilon_{i,t}$, where $g_{i,t}$ is the short-term component and τ_t is the long-term component. In this setting, the long-term component is modeled as a function of macroeconomic variables of interest (e.g., industrial production growth rate and producer price index). We aim at capturing this variation at the daily frequency though.

Following Christensen, Siggaard, and Veliyev (2023), we include four macro-finance variables: the Aruoba, Diebold, and Scotti (2009) business conditions index (ADS), the US 1-month T-bill rate, the Economic Policy Uncertainty (EPU) index from Baker, Bloom, and Davis (2016), and the CBOE implied volatility index (VIX).[7] The ADS index and the 1-month T-bill rate help capture real business activity. The EPU index also is linked to aggregate business activity and stock market activity as it has been shown in prior literature (see, e.g., Pastor and Veronesi 2012; Brogaard and Detzel 2015; Liu and Zhang 2015; and Kang, de Gracia, and Ratti 2017). The VIX index also typically correlates with stock market volatility, so we also evaluate it as a macroeconomic predictor in our empirical exercise.

Furthermore, we incorporate a set of technical indicators for each firm in the sample. Our motivation for testing the ability of technical indicators to predict stock volatility is prior literature. For example, Neely et al. (2014) show that technical indicators similar to those used in our study can forecast the aggregate stock market in the United States, and they outperform a set of macroeconomic variables. He et al. (2021) and Yin and Yang (2016) show that technical indicators help to predict oil prices, both in-sample and out-of-sample. Like Neely et al. (2014), the study by Yin and Yang (2016) shows that technical indicators outperform a set of macroeconomic variables. Finally, Liu and Pan (2020) show that technical indicators improve the accuracy of the stock market volatility forecasts. They find that technical indicators predict better during expansions, while macroeconomic variables predict better during recessions. They conclude that the combination of both technical indicators and macroeconomic variables performs the best. Based on previous literature, it appears justifiable to include an assessment of technical indicators in our empirical analysis.

We consider moving average and momentum technical indicators. Details regarding the way these technical indicators were

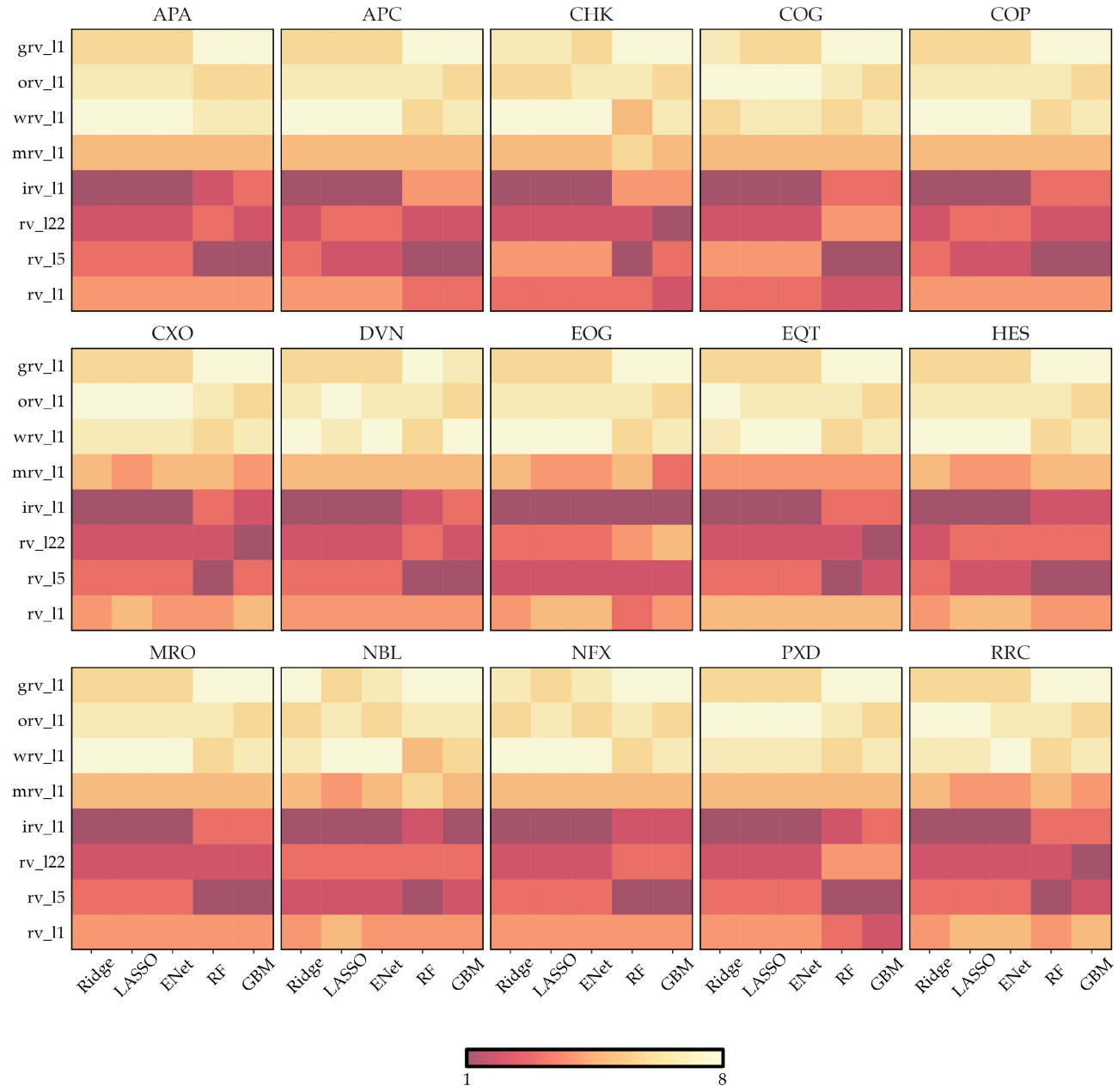


FIGURE 1 | Drivers of firms' realized volatility (original volatility factors).

built are presented in Section 2 above. Thus, we use three sets of predictors in our analysis, the "original" variables (those from Lyócsa and Todorova 2021), the economic variables, and the technical indicators. In this section, to save space, we report results obtained using the original variables, and all the three sets of predictors simultaneously. In the appendix, we show the results for the cases in which the original set of variables is combined only with the macro variables (Table A3), and only with technical indicators (Table A4).

Table 3 shows the results for all predictors variables. The results indicate that the additional set of predictors improves forecasting accuracy for all the firms in the sample. The average improvement in the loss function, as compared with the benchmark model, is 1.90%. However, there are cases in which the ML methods produce reductions in the loss function of around 5.0%. These magnitudes are significant if we compare them with the gains reported by Lyócsa and Todorova (2021) of about 4%. We observe that in this case, in which we have more potential predictors, the best performing model is the forecast combination of the five ML methods used throughout. For the 15 firms in the sample, the reported average gain for this method is 3.6%, with a minimum gain of 1.0% (for the EQT firm) and a maximum gain of 5.50% (for the COP firm). Importantly, the MCS test shows that this forecast combination method belongs to the optimal set for all 15 firms in the sample. In contrast, the benchmark model is never included in this set, which confirms that the ML methods provide more accurate forecasts of the volatility of firms in the oil and gas industry.

The Ridge method is another ML method that yields good results. It is included in the optimal set of models for 13 out of

TABLE 3 | Realized volatility forecasting accuracy of machine learning methods. (Predictors: original volatility factors plus economic variables and technical indicators; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
af-har	0.444	0.432	0.410	0.432	0.570	0.385	0.415	0.455	0.435	0.461	0.427	0.450	0.368	0.412	0.349
ridge	−3.87%*	−3.80%*	−1.12%*	−1.64%	−3.83%	−3.34%*	−1.60%*	−5.32%*	−0.97%*	−4.65%*	−1.88%*	−3.29%*	−1.93%*	−2.91%*	−1.44%*
lasso	−2.97%*	−3.92%	−1.45%*	−1.76%	−3.41%	−2.25%*	−1.83%*	−4.48%	−0.45%*	−3.54%	−1.36%*	−2.90%*	−1.41%	−1.72%	−1.62%*
enet	−3.29%*	−4.21%*	−2.01%*	−1.40%	−3.61%	−2.63%*	−1.90%*	−4.94%*	−0.69%*	−3.62%	−1.52%*	−3.21%*	−2.39%*	−2.23%	−1.63%*
rf	−1.70%	−0.81%	3.16%	−2.63%*	−0.68%	0.94%	2.00%	0.69%	5.13%	3.86%	0.88%	0.01%	3.51%	0.08%	4.52%
gbm	0.84%	−2.33%	2.28%	−1.19%	−3.26%	0.06%	2.64%	−0.67%	1.99%	0.52%	3.17%	−1.78%	0.63%	−1.08%	1.53%
af-har + ridge	−2.92%*	−2.89%	−1.25%*	−2.09%*	−3.04%	−2.61%*	−1.73%*	−3.88%	−1.22%*	−3.63%*	−2.01%*	−2.64%*	−2.07%*	−2.53%	−1.51%*
af-har + lasso	−2.51%*	−2.96%	−1.25%*	−2.11%*	−2.77%	−2.05%*	−1.73%*	−3.48%	−1.06%*	−3.05%	−1.79%*	−2.53%*	−1.75%	−1.93%	−1.42%*
af-har + enet	−2.71%*	−3.05%	−1.53%*	−1.93%*	−2.90%	−2.21%*	−1.81%*	−3.74%	−1.20%*	−3.04%	−1.91%*	−2.64%*	−2.17%*	−2.11%	−1.50%*
af-har + rf	−3.76%*	−3.31%	−1.00%*	−4.06%*	−3.43%	−2.18%*	−1.76%*	−2.56%	0.02%*	−0.92%	−2.66%*	−3.14%	−0.67%	−2.40%	−0.22%
af-har + gbm	−1.67%	−3.17%	−0.70%*	−3.00%*	−4.09%*	−1.93%	−1.15%*	−2.65%	−0.81%*	−1.95%	−1.43%*	−3.18%	−1.69%*	−2.23%	−0.98%*
pooled-avg	−4.46%*	−5.15%*	−2.09%*	−4.23%*	−5.50%*	−3.72%*	−2.67%*	−5.39%*	−1.00%*	−3.79%*	−3.18%*	−5.06%*	−2.48%*	−3.68%*	−1.63%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR +" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

15 firms. In these results, we observe that all the ML estimates, except for the RF and GBM, show some degree of predictability. Only, when tree-based methods are combined with the benchmark model they show some degree of predictability.

We find that the economic variables are better at improving the forecasting ability of estimated models than the technical indicators. See these results in Tables A3 and A4, respectively. Figure 2 shows a heat map with the relative importance of each forecasting variables by individual ML method and firm. A dark (light) red color indicates more (less) importance in the estimation of the forecasting model. Similarly, Lyócsa and Todorova (2021) observe that the volatility factors are significant predictors of a firm's volatility across methods and firms. We find that HAR factors are the most important variables: firms' own lagged volatility (rv_l1), lagged weekly volatility (rv_l5), and lagged monthly volatility (rv_l22). We also identify industry volatility as an important driver of firms' volatility.

In the case of the additional predictors, two variables appear particularly important: the US 1-month Treasury bill ("dgs1mo_l1") when regularization methods are used and the VIX index when tree-based models are used. It is worth noting that the VIX index, which is built using option information, subsumes the stock market volatility factor identified by Lyócsa and Todorova (2021) as a significant predictor. These findings are consistent with prior evidence pointing to the superior informational content of option-implied volatility over past volatility (used to compute measures of RV) for volatility forecasts.

5 | Further Analysis

In this section, we report the results of two additional exercises. First, we provide the findings when we use the MSFE loss function. Second, we show results for the case in which ML methods are estimated using an alternative k-fold cross-validation scheme to set the hyperparameters in ML methods.

5.1 | MSFE Loss Function

We evaluated the alternative ML methods and predictors using the Q-LIKE function, as Patton (2011) recommended. Table 4

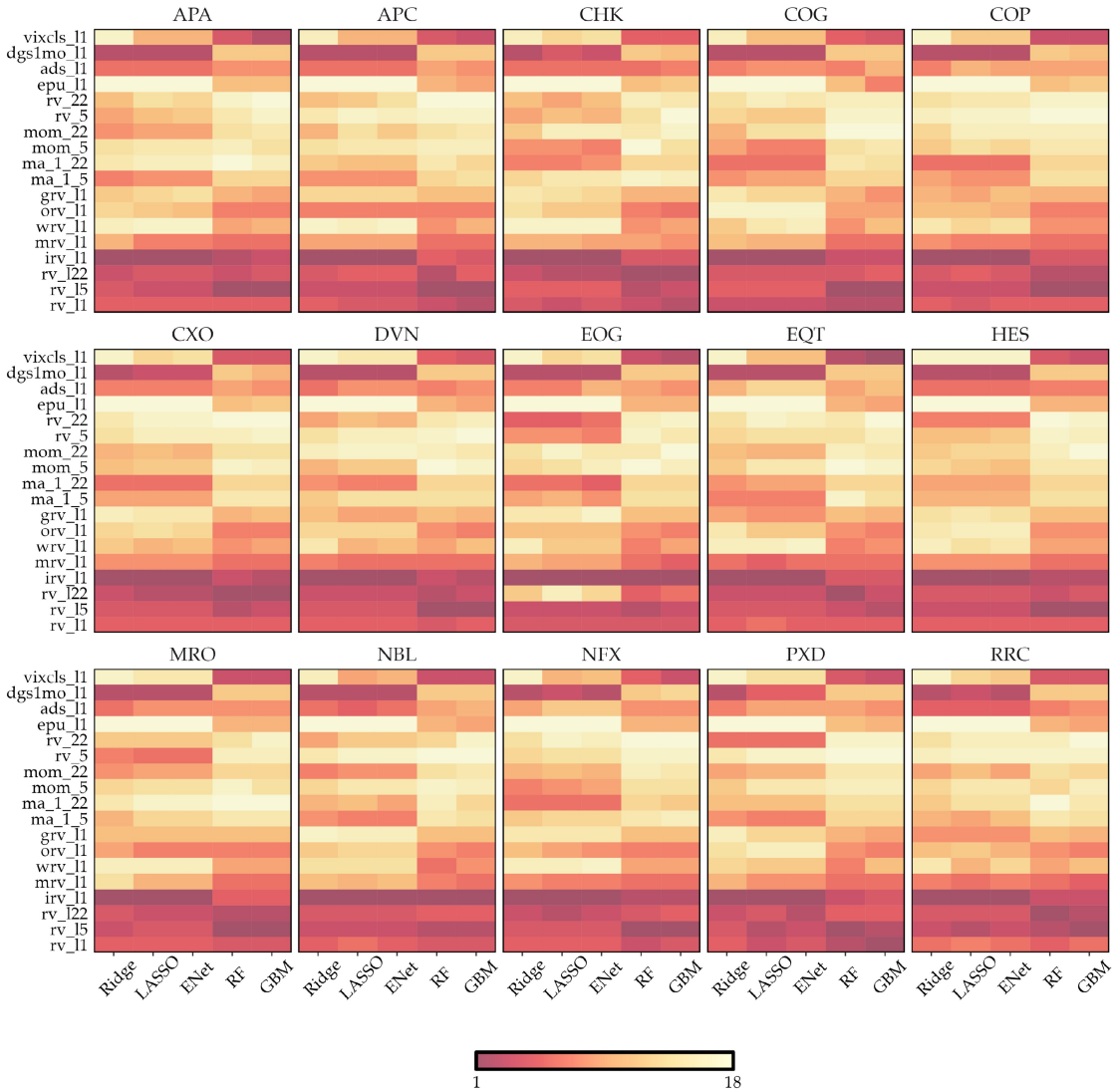


FIGURE 2 | Drivers of firms' volatility (All predictors: original volatility factors, economic variables, and technical indicators).

TABLE 4 | Realized volatility forecasting accuracy of machine learning methods. (Predictors: original volatility factors; MSFE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
af-har	0.394	0.396	0.464*	0.450*	0.428	0.375*	0.375	0.420*	0.382	0.423	0.410	0.417	0.396	0.405	0.348*
ridge	−0.94%*	−0.91%*	−0.76%*	0.70%	−0.88%*	−0.47%*	−0.66%*	−0.44%*	−0.86%*	−1.31%*	−0.54%*	−0.62%	−0.96%*	−0.54%*	−0.43%*
lasso	−0.01%	−0.76%*	−0.59%*	1.09%	−0.57%*	−0.29%*	−0.07%*	−0.27%*	−0.33%*	−1.04%*	−0.23%*	−0.15%	−0.20%	0.41%	−0.57%*
enet	−0.83%*	−0.98%*	−0.61%*	1.02%	−1.03%*	−0.31%*	−0.19%*	−0.51%*	−0.56%*	−1.04%*	−0.42%*	−0.15%	−0.91%*	−0.13%	−0.75%*
rf	1.10%	1.44%	3.23%	0.56%	4.25%	3.79%	3.65%	2.22%	4.28%	4.51%	1.99%	−1.50%*	4.14%	1.68%	6.73%
gbm	2.93%	0.22%*	2.09%	1.02%	0.12%	0.56%	3.82%	1.48%	2.55%	1.83%	3.51%	−2.69%*	1.28%	2.16%	3.62%
af-har + ridge	−0.65%*	−0.60%*	−0.53%*	−0.02%*	−0.59%*	−0.46%*	−0.53%*	−0.51%*	−0.62%*	−0.91%*	−0.66%*	−0.55%	−0.76%*	−0.45%*	−0.53%*
af-har + lasso	−0.16%	−0.50%*	−0.39%*	0.16%*	−0.40%	−0.26%*	−0.15%*	−0.34%*	−0.28%	−0.72%*	−0.24%*	−0.38%	−0.36%	0.03%	−0.48%*
af-har + enet	−0.60%*	−0.68%*	−0.42%*	0.13%*	−0.69%*	−0.34%*	−0.29%*	−0.46%*	−0.43%*	−0.73%*	−0.59%*	−0.44%	−0.74%*	−0.23%	−0.61%*
af-har + rf	−1.91%*	−1.61%*	−0.71%*	−2.00%*	−0.46%	−0.78%*	−0.49%*	−1.27%*	−0.02%	−0.10%	−1.56%*	−3.34%*	0.07%*	−1.23%*	1.22%
af-har + gbm	−0.71%	−1.72%*	−0.59%*	−1.34%*	−1.75%*	−1.41%*	−0.07%*	−1.21%*	−0.27%*	−0.69%*	−0.30%	−3.30%*	−0.80%*	−0.49%	0.48%*
pooled-avg	−1.90%*	−2.21%*	−1.39%*	−1.29%*	−1.79%*	−1.43%*	−0.84%*	−1.75%*	−0.84%*	−1.34%*	−1.50%*	−3.32%*	−1.01%*	−1.16%*	0.00%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the MSFE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR+" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

TABLE 5 | Realized volatility forecasting accuracy of machine learning methods with K-fold cross-validation method. (Predictors: original volatility factors; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
af-har	0.444	0.432	0.410	0.432	0.570	0.385	0.415	0.455	0.435	0.461	0.427	0.450	0.368	0.412	0.349
ridge	−0.75%	−0.86%*	−0.77%*	−0.36%	−1.01%*	−0.54%*	−0.79%*	−0.77%*	−0.73%*	−0.76%*	−0.65%*	−1.11%	−0.99%*	−0.72%*	−0.66%*
lasso	−0.15%	−0.67%*	−0.35%*	−0.08%	−0.22%	−0.36%*	−0.14%	−0.23%	−0.30%*	−0.52%*	−0.26%*	−1.01%	−0.35%	−0.24%	−0.32%*
enet	−0.14%	−0.79%*	−0.41%*	−0.22%	−0.46%	−0.31%*	−0.14%	−0.36%	−0.38%*	−0.51%*	−0.32%*	−0.95%	−0.52%	−0.38%	−0.45%*
rf	1.25%	2.71%	4.53%	0.45%	4.49%	4.47%	4.42%	3.47%	4.87%	5.97%	3.10%	−0.64%	4.51%	2.16%	7.56%
gbm	−0.57%	−1.03%*	2.62%	−0.98%*	0.65%	1.67%	2.27%	0.01%	3.39%	2.40%	2.56%	−2.49%*	1.09%	0.32%	4.21%
af-har + ridge	−0.44%	−0.50%*	−0.46%	−0.23%	−0.58%	−0.33%*	−0.48%	−0.45%	−0.44%*	−0.47%*	−0.42%*	−0.67%	−0.56%	−0.42%	−0.40%*
af-har + lasso	−0.10%	−0.38%	−0.20%	−0.08%	−0.15%	−0.20%*	−0.09%	−0.14%	−0.17%	−0.30%*	−0.16%*	−0.56%	−0.20%	−0.15%	−0.20%*
af-har + enet	−0.11%	−0.44%*	−0.24%	−0.15%	−0.26%	−0.19%*	−0.09%	−0.20%	−0.21%	−0.30%*	−0.20%*	−0.53%	−0.29%	−0.23%	−0.27%*
af-har + rf	−2.02%*	−1.20%	−0.17%	−2.33%*	−0.47%	−0.40%*	−0.18%	−0.85%	0.13%	0.48%	−1.18%*	−3.05%*	0.12%	−1.09%*	1.49%
af-har + gbm	−2.18%*	−2.17%*	−0.49%*	−2.20%*	−1.61%*	−0.98%*	−0.54%*	−1.72%*	0.14%	−0.39%*	−0.69%	−3.34%*	−0.86%*	−1.34%*	0.64%*
pooled-avg	−2.32%*	−2.19%*	−1.04%*	−2.39%*	−1.58%*	−1.18%*	−0.89%*	−1.69%*	−0.55%*	−0.72%*	−1.42%*	−3.57%*	−1.00%*	−1.57%*	0.24%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR+" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

reports forecasting results using the original set of factors but replacing the Q-LIKE loss function with the MSFE loss function. Overall, we observe slightly weaker results, but most of the results remain qualitatively the same. For example, while the benchmark model's forecasts are in the optimal confident set according to the MCS test for 5 of the analyzed firms, the forecast from the model that combines the forecasts of all the ML methods is in the optimal confident set for all but one firm (RRC) in the sample. The average forecasting gain of using the combined pooled model is 1.5%, varying from levels of 0.84% for DVN stock to 2.21% for APC stock. As before, the Ridge method performs the best among the individual ML methods, and tree-based methods work particularly poorly, with the benchmark producing more accurate forecasts. Table A5 shows more robust results, favoring ML methods when we use the complete set of predictors, including economic variables and technical indicators.

5.2 | Alterative Cross-Validation Procedure

Baseline results were computed using a time series cross-validation method described in Section 3.5. As a robustness check, we evaluate the sensitivity of these results to this method and implement an alternative K-fold cross-validation procedure. A K-fold cross-validation in the context of a times series exercise has been considered, for example, by DeMiguel et al. (2023). Table 5 shows the results considering the set of original predictors. We find that using a K-fold cross-validation, which does not consider the time series dimension in the validation stage for tuning the hyperparameters, marginally improved the forecasting ability of the models when compared to the baseline results. However, the baseline results are qualitatively similar, with ML methods outperforming the benchmark model. Table A6 shows the results when we use the complete set of predictors, including economic variables and technical indicators. We observe similar results compared to the time series cross-validation case reported in our baseline results. Besides, we also find that the additional set of predictors significantly improves the forecast accuracy of the ML models.

6 | Conclusions

Forecasting the volatility of oil and gas firms is of particular interest to investors and practitioners trading in the energy spot and derivative markets (see, e.g., Antonakakis et al. 2018; Kang, de Gracia, and Ratti 2017). The previous literature has mainly relied on linear specifications, such as the HAR model, to empirically address this issue. Furthermore, within this literature, volatility factors have been identified as the primary predictors of future volatility at the firm level (see, e.g., Corsi 2009; Lyócsa and Todorova 2021).

In this paper, we study to which extent a set of more flexible ML methods improves the forecasting accuracy of the RV of a representative sample of US firms in the oil and gas industry. Furthermore, we evaluate a set of additional predictors by including economic variables and technical indicators in addition to the volatility factors. Our results indicate that allowing for more flexible models reduces errors in US oil and gas firm RV forecasts by up to an additional 5% as compared with our benchmark model. Individually, the Ridge model performs the best among the set of analyzed models, but the overall winner model is the one that combined forecast for the all the ML methods.

Besides, we find that economic variables such as the US 1-month treasury bill and the option-implied VIX index contain useful information with which to forecast the RV of firms in the US oil and gas industry. The predictive power of the technical indicators is more limited than that of the economic variables we evaluated.

This paper contributes to the small body of literature examining the drivers of US oil and gas firms' RV by evaluating the performance of ML methods that have recently been used to forecast equity volatility at the firm level (see, e.g., Christensen, Siggaard, and Veliyev 2023; Li and Tang 2021; Rahimikia and Poon 2020; Audrino, Sigrist, and Ballinari 2020; Zhu et al. 2023) and at the aggregate level (see, e.g., Mittnik, Robinzonov, and Spindler 2015; Bucci 2020; Díaz, Hansen, and Cabrera 2024; Ma et al. 2023).

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] Neely et al. (2014) show that technical indicators improve forecasts of aggregate U.S. stock market returns more than a set of macroeconomic variables. Liu and Pan (2020) show that technical indicators improve the accuracy of the stock market volatility forecasts. They find that technical indicators predict better during expansions, while macroeconomic variables predict better during recessions. The combination of both technical indicators and macroeconomic variables performs the best. He et al. (2021), Yin and Yang (2016) and Liu, Wang, and Yang (2018) show that technical indicators help to predict oil prices. Similar to Neely et al. (2014), Yin and Yang (2016) find that technical indicators possess in-sample and out-of-sample forecasting power for oil prices, which exceeds that of macroeconomic variables and state-of-the-art oil-macro variables.

[2] See, for example, Ghoddusi, Creamer, and Rafizadeh (2019) for a survey of ML applications in energy economics and finance.

[3] See Ma et al. (2018) for further details regarding the estimation of the technical indicators.

[4] To save space, we have not reported descriptive statistics for the technical indicators of the 15 oil and gas firms in the sample. They are available from the authors upon request.

[5] The industry volatility corresponds to the cross-section volatility of the 15 firms in the Oil and Gas sub-industry; the Market volatility is computed using S&P Mini index futures data; the World market volatility is computed using MSCI World market index data; Oil market volatility is computed using WTI future contracts; and Gas market volatility is calculated using Henry Hub futures contracts.

[6] In the case of the penalized linear regressions (Ridge, Lasso, and Elastic Net), given that all the predictors are normalized before the estimation, we use the absolute value of each method's coefficient estimates as a measure of variable importance. In the case of the tree-based methods, the importance of individual variables was determined based on whether they were selected to split on during the tree building process or not, as well as on the magnitude of the improvement in the squared error (see Breiman et al. 1984 and Friedman, Hastie, and Tibshirani 2010, for more details).

[7] All these predictors are available with a one-day lag, which is adequate for the forecasting framework we are considering, except for the ADS index. The time-series data for the ADS index is updated seven to eight times a month due to the way in which is built by the Federal Reserve Bank of Philadelphia. See details of index construction here: https://www.philadelphiafed.org/-/media/frbp/assets/surveys-and-data/ads/ads-technical-documentation.pdf.

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Appendix A

TABLE A1 | Descriptive statistics.

	Mean	SD	Skew	Kurt	AC1	AC5	AC22
Volatility factors							
Industry	6.88	0.86	0.88	4.03	0.80	0.73	0.61
Market	4.83	1.35	−0.07	3.50	0.59	0.52	0.36
World	4.47	1.15	0.13	3.75	0.75	0.64	0.49
Oil	6.31	1.29	−0.50	3.80	0.42	0.39	0.29
Natgas	7.01	0.99	−0.24	6.12	0.33	0.31	0.15
Macrovariables							
ADS	−0.34	0.79	−2.35	9.23	1.00	0.99	0.86
DGS1MO	0.65	1.30	2.45	7.85	1.00	0.98	0.93
EPU	108.15	67.88	1.79	8.47	0.65	0.54	0.39
VIXCLS	19.91	9.67	2.34	10.29	0.98	0.93	0.81

Note: The table reports descriptive statistics for the volatility factors (the same ones used by Lyócsa and Todorova 2021) and macroeconomic variables. ADS stands for the Aruoba, Diebold, and Scotti (2009) business conditions index. DGS1MO stands for the US 1-month T-bill rate. EPU stands for the Economic Policy Uncertainty index from Baker, Bloom, and Davis (2016) and VIXCLS stands for the implied volatility index (VIX).

TABLE A2 | Correlation matrix for volatility factors and macroeconomic variables.

	Industry	Oil	Market	World	Natgas	epu	ads	dgs1mo	vixcls
Industry	1.000								
Oil	0.480*	1.000							
Market	0.840*	0.430*	1.000						
World	0.870*	0.440*	0.930*	1.000					
Natgas	0.005	0.003	0.022	0.016	1.000				
epu	0.270*	0.190*	0.340*	0.360*	0.066*	1.000			
ads	−0.490*	−0.570*	−0.450*	−0.440*	−0.006	−0.240*	1.000		
dgs1mo	−0.076*	−0.070*	−0.048*	−0.076*	−0.012	−0.220*	−0.021	1.000	
vixcls	0.680*	0.550*	0.730*	0.760*	0.046*	0.450*	−0.630*	−0.100*	1.000

*Statistical significant at 10%.
**Statistical significant at 5%.
***Statistical significant 1%.

TABLE A3 | Realized volatility forecasting accuracy of machine learning methods. (predictors: original volatility factors plus economic variables; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
AF-HAR	0.444	0.432	0.410	0.432	0.570	0.385	0.415	0.455	0.435	0.461	0.427	0.450	0.368	0.412	0.349
ridge	−4.14%*	−4.68%*	−1.61%*	−2.16%	−3.97%*	−4.02%*	−2.08%*	−4.71%*	−1.70%*	−3.66%*	−2.54%*	−3.52%*	−2.35%*	−2.51%*	−2.32%*
lasso	−3.19%	−4.10%	−1.74%*	−2.46%	−3.36%	−3.02%	−2.38%*	−4.29%*	−0.92%*	−2.88%*	−1.75%*	−2.84%*	−1.32%	−1.60%	−2.11%*
enet	−3.74%*	−4.45%	−1.80%*	−2.59%	−3.48%	−3.50%	−2.41%*	−4.33%*	−1.13%*	−2.96%*	−1.58%*	−2.63%*	−1.97%*	−1.70%	−2.37%*
rf	−0.64%	−0.90%	3.33%	−1.60%*	−0.09%	1.89%	1.69%	1.92%	5.85%	5.12%	0.66%	−0.28%	4.20%	0.84%	4.54%
gbm	0.35%	−2.06%	2.43%	−1.38%	−2.27%	1.24%	3.29%	−0.49%	3.71%	0.87%	4.50%	−0.55%	0.51%	0.07%	1.64%
af-har + ridge	−2.70%	−3.09%	−1.24%*	−2.03%*	−2.69%	−2.64%	−1.73%*	−3.13%	−1.38%*	−2.59%*	−2.04%*	−2.47%*	−1.80%*	−1.91%	−1.77%*
af-har + lasso	−2.33%	−2.90%	−1.24%*	−2.27%*	−2.42%	−2.11%	−1.73%*	−3.11%	−1.17%*	−2.30%	−1.74%*	−2.30%*	−1.47%	−1.52%	−1.54%*
af-har + enet	−2.60%	−2.98%	−1.30%*	−2.29%*	−2.49%	−2.38%	−1.79%*	−3.17%	−1.26%*	−2.33%	−1.69%*	−2.24%*	−1.78%*	−1.54%	−1.70%*
af-har + rf	−3.28%*	−3.39%	−0.98%*	−3.59%*	−3.22%	−1.81%	−1.94%*	−2.01%	0.29%*	−0.33%	−2.79%*	−3.24%*	−0.45%	−2.03%*	−0.30%
af-har + gbm	−1.75%	−3.09%	−0.64%	−2.94%*	−3.69%	−1.46%	−1.07%	−2.42%	−0.17%*	−1.66%	−0.70%	−2.80%	−1.62%*	−1.62%	−0.96%
pooled-avg	−4.44%*	−5.46%*	−2.14%*	−4.30%*	−5.27%*	−3.76%*	−2.99%*	−4.67%*	−0.98%*	−2.90%*	−3.01%*	−4.81%*	−2.20%*	−3.15%*	−2.05%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR+" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

TABLE A4 | Realized volatility forecasting accuracy of machine learning methods. (Predictors: original volatility factors plus technical indicators; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
AF-HAR	0.444*	0.432	0.410*	0.432	0.570*	0.385*	0.415*	0.455	0.435*	0.461	0.427*	0.450	0.368	0.412	0.349*
ridge	−1.07%*	−0.86%*	−0.45%*	1.34%	−0.80%*	−0.46%*	−0.34%*	−2.04%*	0.11%*	−2.44%*	−0.44%*	−0.62%	−0.70%*	−1.35%*	0.40%*
lasso	−0.10%	−1.13%*	−0.49%*	2.30%	−0.50%*	0.02%*	0.24%*	−1.38%	0.21%*	−1.60%*	0.29%*	−0.44%	−0.68%*	−0.77%*	−0.07%*
enet	−0.78%*	−1.11%*	−0.62%*	1.82%	−0.77%*	−0.20%*	0.20%*	−1.72%*	0.13%*	−1.98%*	−0.23%*	−0.73%	−1.25%*	−0.90%*	−0.06%*
rf	1.16%	1.82%	3.16%	−0.40%	2.82%	2.89%	3.87%	1.76%	4.20%	4.85%	2.95%	−0.77%	2.94%	1.70%	6.86%
gbm	1.04%	−0.12%*	0.84%	1.17%	−0.08%*	1.73%	3.10%	0.26%	2.80%	2.80%	4.53%	−3.92%*	1.13%	0.20%	3.58%
af-har + ridge	−1.22%*	−1.01%*	−0.63%*	−0.13%	−1.16%*	−0.84%*	−0.78%*	−1.84%*	−0.38%*	−2.05%*	−0.91%*	−1.02%	−1.25%*	−1.39%*	−0.32%*
af-har + lasso	−0.65%	−0.99%*	−0.56%*	0.39%	−0.71%*	−0.50%*	−0.37%*	−1.37%	−0.20%*	−1.56%	−0.41%*	−0.78%	−0.95%	−0.97%*	−0.35%*
af-har + enet	−0.98%*	−1.02%*	−0.62%*	0.17%	−0.94%*	−0.67%*	−0.47%*	−1.53%	−0.31%*	−1.70%*	−0.74%*	−0.93%	−1.31%*	−1.06%	−0.39%*
af-har + rf	−1.99%*	−1.63%*	−0.75%*	−2.72%*	−1.23%*	−1.16%*	−0.35%*	−1.61%	−0.10%*	−0.11%	−1.23%*	−3.04%*	−0.59%*	−1.30%*	1.23%*
af-har + gbm	−1.33%*	−2.22%*	−1.11%*	−1.53%*	−1.98%*	−1.01%*	−0.46%*	−1.80%	−0.16%*	−0.47%*	−0.26%*	−3.80%*	−1.02%*	−1.33%*	0.43%*
pooled-avg	−2.16%*	−2.55%*	−1.59%*	−1.26%	−2.06%*	−1.36%*	−0.84%*	−2.99%*	−0.37%*	−1.87%*	−1.30%*	−3.70%*	−1.61%*	−2.01%*	0.36%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR +" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

TABLE A5 | Realized volatility forecasting accuracy of machine learning methods. (Predictors: original volatility factors plus economic variables and technical indicators; MSFE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
AF-HAR	0.394	0.396	0.464	0.450	0.428	0.375	0.375	0.420	0.382*	0.423	0.410	0.417	0.396	0.405	0.348
ridge	−3.08%*	−3.40%*	−0.40%*	−1.83%	−3.73%*	−2.76%*	−1.54%*	−4.03%*	−0.50%*	−4.22%*	−1.75%*	−2.89%*	−1.25%*	−2.35%*	−0.81%*
lasso	−2.28%	−3.24%*	−1.03%*	−1.81%	−3.08%	−1.70%*	−1.55%*	−3.10%	−0.19%*	−3.02%	−1.31%	−2.43%*	−0.88%	−1.30%	−0.87%*
enet	−2.99%*	−3.69%*	−1.39%*	−1.81%	−3.51%	−2.07%*	−1.68%*	−3.65%*	−0.49%*	−3.13%	−1.48%	−2.69%*	−1.92%*	−1.77%	−0.91%*
rf	−1.22%	−0.15%	3.30%	−1.63%*	−0.37%	0.60%	1.70%	0.73%	4.85%	3.57%	0.49%	−0.44%	3.45%	0.41%	4.67%
gbm	1.40%	−1.28%	2.43%	−1.04%	−2.33%	0.43%	2.12%	−0.29%	2.26%	1.18%	3.28%	−2.02%	1.05%	−0.74%	2.05%
af-har + ridge	−2.50%*	−2.72%*	−0.88%*	−2.10%*	−3.03%	−2.32%*	−1.70%*	−3.20%*	−0.97%*	−3.35%*	−1.88%*	−2.42%*	−1.75%*	−2.20%*	−1.19%*
af-har + lasso	−2.10%	−2.65%	−1.00%*	−2.08%*	−2.61%	−1.76%*	−1.59%*	−2.78%	−0.89%*	−2.71%	−1.63%*	−2.25%*	−1.44%	−1.67%	−1.03%*
af-har + enet	−2.49%*	−2.80%*	−1.18%*	−2.06%*	−2.86%	−1.93%*	−1.70%*	−3.07%*	−1.04%*	−2.71%	−1.78%*	−2.35%*	−1.89%*	−1.83%	−1.13%*
af-har + rf	−3.33%*	−2.75%	−0.85%*	−3.34%*	−3.13%	−2.33%*	−1.83%*	−2.32%	0.01%*	−0.90%	−2.66%*	−3.25%	−0.57%*	−2.13%*	−0.07%*
af-har + gbm	−1.36%	−2.61%	−0.68%*	−2.73%*	−3.55%	−1.87%*	−1.33%*	−2.33%	−0.63%*	−1.58%	−1.19%	−3.31%	−1.41%*	−2.04%*	−0.70%*
pooled-avg	−3.85%*	−4.44%*	−1.72%*	−3.97%*	−5.09%*	−3.44%*	−2.70%*	−4.40%*	−0.77%*	−3.37%*	−2.96%*	−4.88%*	−2.00%*	−3.22%*	−1.09%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the MSFE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR+" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.

TABLE A6 | Realized volatility forecasting accuracy of machine learning methods with K-fold cross-validation method . (Predictors: original volatility factors plus economic variables and technical indicators; Q-LIKE loss function).

	APA	APC	CHK	COG	COP	CXO	DVN	EOG	EQT	HES	MRO	NBL	NFX	PXD	RRC
AF-HAR	0.444	0.432	0.410	0.432	0.570	0.385	0.415	0.455	0.435	0.461	0.427	0.450	0.368	0.412	0.349
ridge	−3.27%*	−3.61%	−1.02%*	−3.57%	−4.09%	−3.76%*	−1.58%*	−5.14%*	−1.07%*	−4.26%*	−2.02%	−3.05%	−2.42%*	−3.62%*	−1.17%*
lasso	−2.99%*	−3.69%	−0.89%*	−3.32%	−3.26%	−3.26%	−1.07%*	−4.85%*	−1.10%*	−3.34%	−1.75%	−2.89%*	−2.34%*	−3.18%	−0.94%*
enet	−3.02%*	−3.61%	−0.83%*	−3.27%	−3.58%	−3.26%	−1.19%*	−4.92%*	−1.27%*	−3.56%	−1.90%	−3.06%	−2.50%*	−3.23%	−0.83%*
rf	−2.12%	−1.90%	1.09%	−2.86%*	−1.83%	−0.32%	0.62%	−0.82%	3.64%	2.19%	−0.05%	−2.11%	1.61%	−0.99%	2.98%
gbm	−0.88%	−3.78%*	1.22%	−3.85%*	−3.47%	−1.33%	0.27%	−2.15%	2.01%	−0.01%	0.27%	−2.75%	0.08%	−1.94%	1.78%
af-har + ridge	−2.60%*	−2.82%	−1.15%*	−2.84%	−3.11%	−2.90%	−1.63%*	−3.82%	−1.19%*	−3.42%*	−2.02%	−2.55%	−2.02%*	−2.77%	−1.25%*
af-har + lasso	−2.46%*	−2.85%	−0.96%*	−2.73%	−2.73%	−2.61%	−1.33%	−3.65%	−1.18%*	−3.03%	−1.85%	−2.45%	−1.96%*	−2.49%	−1.02%*
af-har + enet	−2.46%*	−2.80%	−0.93%*	−2.69%	−2.86%	−2.60%	−1.39%*	−3.66%	−1.28%*	−3.11%	−1.91%	−2.51%	−2.03%*	−2.50%	−0.97%*
af-har + rf	−3.62%*	−3.56%	−1.69%*	−3.93%*	−3.65%	−2.46%	−2.11%*	−2.97%	−0.44%*	−1.43%	−2.78%*	−3.83%	−1.34%	−2.62%	−0.73%*
af-har + gbm	−2.79%*	−4.11%	−1.43%*	−4.12%*	−4.20%*	−2.79%	−2.09%*	−3.23%	−0.79%*	−2.00%	−2.57%*	−3.89%	−1.84%	−2.79%	−0.86%*
pooled-avg	−4.51%*	−5.30%*	−2.15%*	−5.37%*	−5.46%*	−4.53%*	−2.79%*	−5.68%*	−1.27%*	−3.82%*	−3.52%*	−5.37%*	−2.85%*	−4.40%*	−1.30%*

Note: The table reports forecasting performance values for ML models' one-day ahead predictions of firms' realized volatility. Original volatility factors refer to those in Lyócsa and Todorova (2021). It shows the value of the Q-LIKE loss function for the benchmark model (AF-HAR) and the relative percentual change with respect to this benchmark model. The * symbol indicates that a model belongs to the superior set of forecasting models according to the MCS test of Hansen, Lunde, and Nason (2011). The model labeled as "AF-HAR +" refers to a forecast obtained by combining the benchmark model and a given ML model. All combination forecasts were weighted equally. The "Pooled-avg" row shows the performance of a model that (equally) combined the five ML methods under consideration.